

Causal Discovery from Complex Survey Data

Theory (main)

1 Theoretical Properties

This section establishes the large-sample properties of the swa-oBN. We show that under standard survey design conditions and the ordinal regression specification of Ni and Mallick (2022), the survey-weighted pseudo-BIC consistently selects the true data-generating DAG — including full edge orientation within Markov equivalence classes. All proofs are given in Appendix A.

1.1 Asymptotic Framework

The analysis requires a joint asymptotic framework that accounts for two sources of randomness: the superpopulation generating the finite population, and the sampling design selecting units from it.

Let ξ denote a superpopulation model with true causal DAG $\mathcal{G}^*$ and parameters θ^* . For each population size N , a finite population $U_N = \{1, \dots, N\}$ is generated as N independent draws from ξ . From U_N , a probability sample S_n of size $n = n(N)$ is drawn according to a sampling design $p_N(S)$ with known first-order inclusion probabilities $\pi_i = \Pr(i \in S_n)$ and survey weights $w_i = 1/\pi_i$. We require $N \rightarrow \infty$ and $n \rightarrow \infty$ with sampling fraction $n/N \rightarrow f \in [0, 1)$. All convergence statements are under the joint (superpopulation $\times$ design) probability.

1.2 Assumptions

We organize the assumptions into four groups: causal, parametric, survey design, and regularity.

1.2.1 Causal Assumptions

Assumption 1 (Causal Sufficiency). *All common causes of any pair (X_j, X_k) in $\mathbf{X} = (X_1, \dots, X_q)$ are included in the measured variables $\mathbf{X} \cup \mathbf{Z}$.*

Assumption 2 (Causal Markov Property). *The superpopulation distribution $P_\xi(\mathbf{X})$ factorizes with respect to the true causal DAG $\mathcal{G}^*$:*

$$P_\xi(\mathbf{X}) = \prod_{j=1}^q P_\xi(X_j \mid X_{\text{pa}(j)}).$$

Assumption 3 (Faithfulness). *Every conditional independence in P_ξ is entailed by the Markov property of $\mathcal{G}^*$. That is, $X_A \perp\!\!\!\perp X_B \mid X_C$ in P_ξ if and only if A and B are d -separated by C in $\mathcal{G}^*$.*

1.2.2 Parametric Assumptions

Let $X_j \in \{1, \dots, L_j\}$ with $L_j \geq 3$ for each $j = 1, \dots, q$.

Assumption 4 (Ordinal Regression Specification). *Each conditional distribution under the true DAG $\mathcal{G}^*$ follows a cumulative link model:*

$$P_\xi(X_j \leq \ell \mid X_{\text{pa}(j)}, \mathbf{Z}) = F\left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \delta'_j \mathbf{Z} - \alpha_j\right), \quad \ell = 1, \dots, L_j,$$

where F is a known, strictly increasing link function (e.g., probit), $\gamma_{j1} < \dots < \gamma_{j,L_j} = \infty$ are threshold parameters with $\gamma_{j1} = 0$, β_{jk} are parent effect parameters with $\beta_{jk,1} = 0$, δ_j are covariate adjustment parameters, and α_j is an intercept. The covariate vector $\mathbf{Z}$ is not a node in the DAG and enters every node's conditional identically across candidate DAGs.

Remark 1. *We restrict the state space to $L_j \geq 3$ because the unique identifiability of the ordinal Bayesian network relies on the asymmetry of the cumulative link function across multiple thresholds. If binary nodes ($L_j = 2$) are present, the conditional model collapses to a standard binary regression, and causal directionality between two binary nodes is only identifiable up to the Markov equivalence class unless additional structural assumptions are made.*

Assumption 5 (Ordinal Identifiability). *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be any two Markov-equivalent DAGs, and let θ_1^* denote the true parameter under $\mathcal{G}_1 = \mathcal{G}^*$. Define the pseudo-true parameter under $\mathcal{G}_2$ as $\theta_2^\dagger = \arg \max_\theta \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}_1, \theta_1^*) \log P(\mathbf{x} \mid \mathcal{G}_2, \theta)$. Then $P(\mathbf{X} \mid \mathcal{G}_1, \theta_1^*) \not\equiv P(\mathbf{X} \mid \mathcal{G}_2, \theta_2^\dagger)$.*

Assumption 5 is the central result of Ni and Mallick (2022), who proved it for the model without covariates. The following proposition extends it to the covariate-adjusted case.

Proposition 1 (Identifiability under Covariate Adjustment). *If Assumption 5 holds for the oBN model without covariates, it holds for the covariate-adjusted model in Assumption 4.*

Proof sketch. For each fixed value $\mathbf{Z} = \mathbf{z}$, the covariate-adjusted model reduces to an oBN with shifted intercepts $\tilde{\alpha}_j(\mathbf{z}) = \alpha_j + \delta'_j \mathbf{z}$. The identifiability argument of Ni and Mallick (2022) operates on the functional form of the cumulative link — specifically, the asymmetry between $P(X_j | X_k)$ and $P(X_k | X_j)$ under ordinal regression — which depends on the threshold structure and parent effect parameters, not on the intercept value. Shifting the intercept by $\delta'_j \mathbf{z}$ preserves this asymmetry for each $\mathbf{z}$. The KL divergence $\text{KL}(P(\mathbf{X} | \mathcal{G}_1, \theta_1^*, \mathbf{z}) \| P(\mathbf{X} | \mathcal{G}_2, \theta_2^*(\mathbf{z}), \mathbf{z})) > 0$ for each $\mathbf{z}$, and hence for the marginal (averaged over $\mathbf{Z}$). $\square$

Assumption 6 (Non-Degeneracy). $P_\xi(X_j = \ell | X_{\text{pa}(j)} = x_{\text{pa}(j)}, \mathbf{Z} = \mathbf{z}) > 0$ for all j , all levels ℓ , all parent configurations $x_{\text{pa}(j)}$, and all covariate values $\mathbf{z}$ in the support of $\mathbf{Z}$.

1.2.3 Survey Design Assumptions

Assumption 7 (Positivity). Every unit $i \in U_N$ has a known, strictly positive first-order inclusion probability: $\pi_i = \Pr(i \in S_n) > 0$.

Assumption 8 (Bounded Weights). There exist constants $0 < c \leq C < \infty$, independent of N , such that $c \leq n\pi_i/(n/N)^{-1} \leq C$ for all $i \in U_N$. Equivalently, $c \leq nw_i/N \leq C$, so that no single unit dominates the pseudo-likelihood.

Assumption 9 (Design Consistency). For any bounded, measurable function g on the sample space of $\mathbf{X}$,

$$\frac{1}{N} \sum_{i \in S_n} w_i g(\mathbf{x}_i) - \frac{1}{N} \sum_{i \in U_N} g(\mathbf{x}_i) \xrightarrow{p} 0$$

under the design distribution p_N , for each fixed realization of U_N .

Assumption 9 is satisfied by standard survey designs (stratified random sampling, probability-proportional-to-size sampling, and multi-stage designs with a growing number of primary sampling units) under Assumptions 7–8; see (?, Ch. 1) and (?).

1.2.4 Regularity Assumptions

Assumption 10 (Regularity). For each DAG $\mathcal{G}$: (i) the parameter space $\Theta_{\mathcal{G}}$ is compact; (ii) $\log P(\mathbf{x} | \mathcal{G}, \theta, \mathbf{z})$ is twice continuously differentiable in θ for each $(\mathbf{x}, \mathbf{z})$; (iii) the superpopulation expected log-likelihood $Q_\xi(\theta | \mathcal{G}) = \mathbb{E}_\xi[\log P(\mathbf{X} | \mathcal{G}, \theta, \mathbf{Z})]$ has a unique maximizer $\theta_{\mathcal{G}}^*$ with nonsingular Hessian; and (iv) the family $\{\theta \mapsto \log P(\mathbf{x} | \mathcal{G}, \theta, \mathbf{z}) : (\mathbf{x}, \mathbf{z}) \in \text{supp}(P_\xi)\}$ satisfies a Lipschitz condition in θ uniformly over $(\mathbf{x}, \mathbf{z})$.

For the oBN with finitely many ordinal levels, condition (iv) is automatic since the sample space of $\mathbf{X}$ is finite and F is smooth.

1.3 Main Results

Define the survey-weighted pseudo-log-likelihood under DAG $\mathcal{G}$ as

$$\tilde{\ell}(\theta \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i),$$

the pseudo-MLE as $\hat{\theta}_{\mathcal{G}}^w = \arg \max_{\theta} \tilde{\ell}(\theta \mid \mathcal{G})$, and the survey-weighted pseudo-BIC as

$$\widetilde{\text{BIC}}(\mathcal{G}) = -2 \tilde{\ell}(\hat{\theta}_{\mathcal{G}}^w \mid \mathcal{G}) + K_{\mathcal{G}} \cdot \rho_n,$$

where $K_{\mathcal{G}} = \sum_{j=1}^q (L_j - 1 + \sum_{k \in \text{pa}(j)} (L_k - 1) + \dim(\delta_j))$ is the number of free parameters and ρ_n is a penalty sequence satisfying $\rho_n \rightarrow \infty$ and $\rho_n/N \rightarrow 0$.

Proposition 2 (Pseudo-MLE Consistency). *Under Assumptions 4 and 6–10, for each DAG $\mathcal{G}$,*

$$\hat{\theta}_{\mathcal{G}}^w \xrightarrow{p} \theta_{\mathcal{G}}^*$$

under the joint (superpopulation $\times$ design) probability, where $\theta_{\mathcal{G}}^$ is the superpopulation KL-optimal parameter under $\mathcal{G}$.*

Proof sketch. The argument chains two convergences. First, by Assumption 9, the normalized pseudo-log-likelihood $N^{-1} \tilde{\ell}(\theta \mid \mathcal{G})$ converges in probability to the normalized census log-likelihood $N^{-1} \ell_U(\theta \mid \mathcal{G}) = N^{-1} \sum_{i \in U_N} \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i)$ uniformly in θ , by a Glivenko–Cantelli argument for the function class $\{\theta \mapsto \log P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z})\}$ under bounded weights (Assumption 8). Second, by the superpopulation law of large numbers (Assumption 6), the normalized census log-likelihood converges almost surely to $Q_{\xi}(\theta \mid \mathcal{G})$. The unique maximizer property (Assumption 10(iii)) and the compactness of $\Theta_{\mathcal{G}}$ then give convergence of the argmax. The full proof is in Appendix A.1. $\square$

Theorem 1 (DAG Selection Consistency — Within Markov Equivalence Classes). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true data-generating DAG and $\mathcal{G}'$ any DAG Markov-equivalent to $\mathcal{G}^*$ with $\mathcal{G}' \neq \mathcal{G}^*$. Then*

$$\Pr(\widetilde{\text{BIC}}(\mathcal{G}^*) < \widetilde{\text{BIC}}(\mathcal{G}')) \rightarrow 1.$$

Proof sketch. Since $\mathcal{G}^*$ and $\mathcal{G}'$ are Markov-equivalent, they share the same skeleton and hence the same number of parameters: $K_{\mathcal{G}^*} = K_{\mathcal{G}'}$. The penalty terms cancel in the BIC difference, which reduces to

$$\widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}') = -2 [\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w \mid \mathcal{G}')].$$

By Proposition 2 and the uniform convergence established in its proof,

$$\begin{aligned} \frac{1}{N} [\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w \mid \mathcal{G}')] &\xrightarrow{p} Q_{\xi}(\theta_{\mathcal{G}^*}^* \mid \mathcal{G}^*) - Q_{\xi}(\theta_{\mathcal{G}'}^{\dagger} \mid \mathcal{G}') \\ &= \text{KL}(P_{\xi}(\cdot \mid \mathcal{G}^*, \theta^*) \parallel P(\cdot \mid \mathcal{G}', \theta^{\dagger})). \end{aligned}$$

By Assumption 5 (ordinal identifiability, extended to the covariate-adjusted case by Proposition 1), this KL divergence is strictly positive. Therefore the pseudo-log-likelihood ratio diverges as $N \cdot \text{KL} \cdot (1 + o_p(1)) \rightarrow \infty$, and $\widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}') \rightarrow -\infty$ in probability. The full proof is in Appendix A.2. $\square$

This is the key result: the survey-weighted oBN can correctly orient edges within Markov equivalence classes, a property that neither categorical BNs nor constraint-based methods possess for ordinal data. Importantly, this result holds for *any* penalty sequence ρ_n satisfying the stated conditions, because the penalty cancels entirely — the discrimination within equivalence classes is driven solely by the pseudo-likelihood, not the penalty.

Theorem 2 (DAG Selection Consistency — Across Markov Equivalence Classes). *Under Assumptions 1–10, let $\mathcal{G}''$ be any DAG not Markov-equivalent to $\mathcal{G}^*$. Then:*

- (a) *If $\mathcal{G}''$ has strictly fewer edges than $\mathcal{G}^*$ (underfitting), $\Pr(\widetilde{\text{BIC}}(\mathcal{G}^*) < \widetilde{\text{BIC}}(\mathcal{G}'')) \rightarrow 1$.*
- (b) *If $\mathcal{G}''$ has strictly more edges than $\mathcal{G}^*$ (overfitting) and the penalty satisfies $\rho_n/N \rightarrow 0$ and $\rho_n \rightarrow \infty$, then $\Pr(\widetilde{\text{BIC}}(\mathcal{G}^*) < \widetilde{\text{BIC}}(\mathcal{G}'')) \rightarrow 1$.*

Proof sketch. For part (a), the underfitting DAG $\mathcal{G}''$ misses at least one true edge. By faithfulness (Assumption 3), it encodes a conditional independence that does not hold in P_{ξ} , so $Q_{\xi}(\theta_{\mathcal{G}^*}^* \mid \mathcal{G}^*) > Q_{\xi}(\theta_{\mathcal{G}''}^{\dagger} \mid \mathcal{G}'')$. The pseudo-log-likelihood ratio diverges at rate N , overwhelming any penalty difference.

For part (b), the overfitting DAG $\mathcal{G}''$ includes at least one spurious edge, so $K_{\mathcal{G}''} > K_{\mathcal{G}^*}$. The pseudo-log-likelihood gain from the spurious parameters is $O_p(1)$ (by a Wilks-type argument for the pseudo-likelihood), while the penalty difference $(K_{\mathcal{G}''} - K_{\mathcal{G}^*})\rho_n \rightarrow \infty$. Therefore the BIC favors the sparser $\mathcal{G}^*$. The full proof is in Appendix A.3. $\square$

1.4 The Role of n_{eff} and Penalty Calibration

Theorems 1 and 2 establish consistency for any penalty ρ_n satisfying the sandwich condition $\rho_n \rightarrow \infty$, $\rho_n/N \rightarrow 0$. In practice, the choice of ρ_n affects finite-sample model selection. We use $\rho_n = \log(n_{\text{eff}})$ with Kish’s effective sample size $n_{\text{eff}} = (\sum w_i)^2 / \sum w_i^2$, following the information-theoretic justification of (?).

The motivation is as follows. The pseudo-log-likelihood $\tilde{\ell}(\theta \mid \mathcal{G})$ is of order N (it estimates the census log-likelihood), but the statistical precision of $\hat{\theta}_{\mathcal{G}}^w$ is governed by the design-based variance,

which is of order $1/n_{\text{eff}}$. When the BIC is used for model selection, the penalty must be calibrated to the scale of the pseudo-likelihood ratio under the null (no true edge). Under standard asymptotics, this ratio is

$$\Lambda_w = 2 \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}''}^w \mid \mathcal{G}'') - \tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) \right] \xrightarrow{d} \sum_{j=1}^{K_{\mathcal{G}''} - K_{\mathcal{G}^*}} \delta_j \chi_{1,j}^2,$$

where δ_j are the generalized design effects — eigenvalues of $\mathcal{J}_{\xi}^{-1} V_p$, with $\mathcal{J}_{\xi}$ the superpopulation Fisher information and V_p the design-based variance of the pseudo-score. The mean of Λ_w under the null is $\sum_j \delta_j$, which is approximately $(K_{\mathcal{G}''} - K_{\mathcal{G}^*}) \cdot \bar{\delta}$, where $\bar{\delta}$ is the average design effect. Using $\rho_n = \log(n_{\text{eff}})$ with the Kish approximation $\bar{\delta} \approx n/n_{\text{eff}}$ ensures that the penalty grows at the rate appropriate to the effective information content of the sample.

Two observations are important for practice. First, for discrimination within Markov equivalence classes (Theorem 1), the penalty plays no role, so the choice of n_{eff} does not affect the oBN’s distinctive contribution — full edge orientation. Second, the Kish n_{eff} is a global scalar approximation to a matrix-valued, node-specific quantity. When design effects are heterogeneous across node regressions, this approximation may lead to modest over- or under-penalization of specific edges in finite samples. A node-specific correction, replacing n_{eff} with $\hat{n}_{\text{eff},j}$ estimated from the ratio of the pseudo-score variance to the Hessian for each node’s ordinal regression, would provide tighter calibration at the cost of additional computation.

References

Ni, Y., and Mallick, B. (2022), “Ordinal causal discovery,” in *Proceedings of the thirty-eighth conference on uncertainty in artificial intelligence*, PMLR, pp. 1530–1540.